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Editors
Journal of Global Optimization

Dear Editors,

Please consider "Finite Menus of Spectral Subspaces: Realizability and the Cost of Convexification" as an original research article.

The paper studies the global minimax design of a fixed number of orthogonal subspace presets over a scalar resource-price interval. Its central question is when an approximation based only on the fully adaptive value curve can actually be implemented by feasible presets. We give an exact criterion on every price subinterval in terms of available slopes on optimal faces, express that criterion through spectral cutoff eigenspaces, and bound the additional presets needed to repair a convexified menu by the number of exceptional prices. That number is finite for each fixed spectral instance.

The contribution is a structural refinement of established representative-set, scalarization, and finite-adaptability ideas. We explicitly compare their assumptions and conclusions with ours. We do not claim a new general minimax duality, greedy covering principle, or spectral perturbation theorem. Synthetic computational evidence supports the construction with independently replayable continuous-price upper and lower bounds, including banded instances of dimension 4096. All declared slowdowns, limitations, and development failures are retained; the paper makes no validated physical-application claim.

We believe these questions fit the journal's theoretical and computational global optimization scope. The source and evidence artifact accompanies the manuscript. Substantive AI assistance is disclosed in the manuscript.

Draft: final submission confirmation pending. The sole author's identity, funding, competing interests, and contributions have been supplied. Before sending this letter, confirm approval of the final files and the publication and concurrent-submission status, and insert the verified public artifact address. No external mathematical review has yet been obtained.

Sincerely,

Jevint Felixciano
Independent researcher
jevintfelixciano@protonmail.com